

Islamic University of Technology

Department of Computer Science and Engineering (CSE)
B.Sc. in Software Engineering

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Project Proposal
Plastic_box_solver

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Plastic_box_solver

1 Project Overview

Plastic_box_solver is a web-based Rubik's Cube assistant that lets users scan their scrambled cube using a phone camera, then either solve it themselves on an interactive 3D model or follow an animated step-by-step guide powered by the Kociemba algorithm. The app runs entirely in the browser, tracks solve time and efficiency, and maintains a global leaderboard so users can compare their progress.

2 Motivation Behind the Project

Most people who own a Rubik's Cube never learn to solve it because available resources — video tutorials or raw algorithm sheets — are difficult to follow alongside a physical cube. A browser-based tool that scans the cube, renders it in 3D, and walks the user through each move interactively removes that barrier entirely. Beyond learning, competitive solvers have a genuine need to measure their efficiency against the optimal solution; having a live timer, move counter, and leaderboard turns a solitary puzzle into an engaging, trackable activity.

3 User Requirement Analysis

Stakeholder

One of our friends, who recently got into Rubik's Cube solving, mentioned that he constantly struggles with following YouTube tutorials while also handling his physical cube. He expressed that he would find it far more convenient if an app could simply scan his cube and show him, on an animated model, exactly which moves to make next — without having to switch back and forth between a video and the physical cube.

Functional Requirements

- The system shall allow users to register, log in, and maintain a personal profile.
- The system shall accept a scrambled cube state via camera scan or manual color input.
- The system shall detect and validate all 54 sticker colors from the submitted images.
- The system shall render the cube as a fully interactive, rotatable 3D model in the browser.
- The system shall compute an optimal solution using the Kociemba two-phase algorithm for auto-solve function.
- The system shall support an Interactive Solve mode where users manipulate the cube themselves.
- The system shall support a Guided Solve mode with animated move-by-move instructions that follows traditional speed cube solving methods.

- The system shall track solve time and move count to calculate efficiency percentage per session.
- The system shall persist user history that the user can see in details along with their personal best.
- The system shall display a global leaderboard along with the rank of the user.

Non-Functional Requirements

- Color detection shall be accurate under standard indoor lighting.
- Solution computation shall return a result within two seconds.
- The application shall be responsive and usable on mobile browsers without installation. (This doesn't follow SRS NFR guideline, can't test this requirement. Might wanna change it.)

4 Flow Chart / Class Diagram

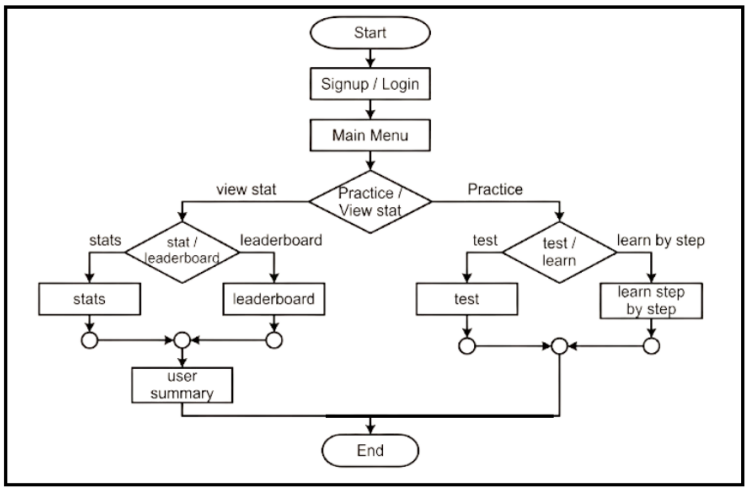


Figure 1: Application Flow Chart

5 Key Features

Cube Input

The user scans each face of their cube using a phone camera or webcam. The Python backend processes each image with OpenCV to detect the nine sticker colors per face and returns a 2D preview for confirmation. If lighting conditions make detection unreliable, a manual click-input interface lets the user set each sticker color directly on a flat 2D cube net.

Interactive 3D Cube

The cube is rendered in the browser using React Three Fiber, built from 26 individual cubie meshes. OrbitControls let the user drag to rotate the view freely. Face rotations triggered by on-screen buttons (U, R, F, etc.) are smoothly animated, and the 3D model stays in sync with the logical cube state at all times.

Solve Modes

In Interactive Mode, the user solves the cube themselves; a timer and move counter run live, and the optimal move count is shown on completion for comparison. In Guided Mode, the CFOP move list appears in a sidebar and the user can step through moves one at a time or trigger Auto-Solve to watch the full solution animate automatically with can follow both CFOP and Kociemba algorithms based on the user's preference.

Statistics and Leaderboard

The sidebar shows current solve time, moves taken, and efficiency percentage in real time. All completed solves are saved to the database. Users can view their personal best and compare against others on a global leaderboard ranked by solve time and efficiency.

6 Tools and Technologies

Technology Stack

Table 1: List of Tools and Technologies

Category	Tool / Technology
Frontend Framework	React + Vite
3D Rendering	React Three Fiber, @react-three/drei
Styling	Tailwind CSS
State Management	Zustand
HTTP Client	Axios
Backend Framework	FastAPI (Python)
Computer Vision	OpenCV + NumPy
Solving Algorithm	kociemba
ORM	SQLAlchemy
Database	PostgreSQL
API Testing	Postman
Version Control	Git & GitHub

7 Proposed Timeline

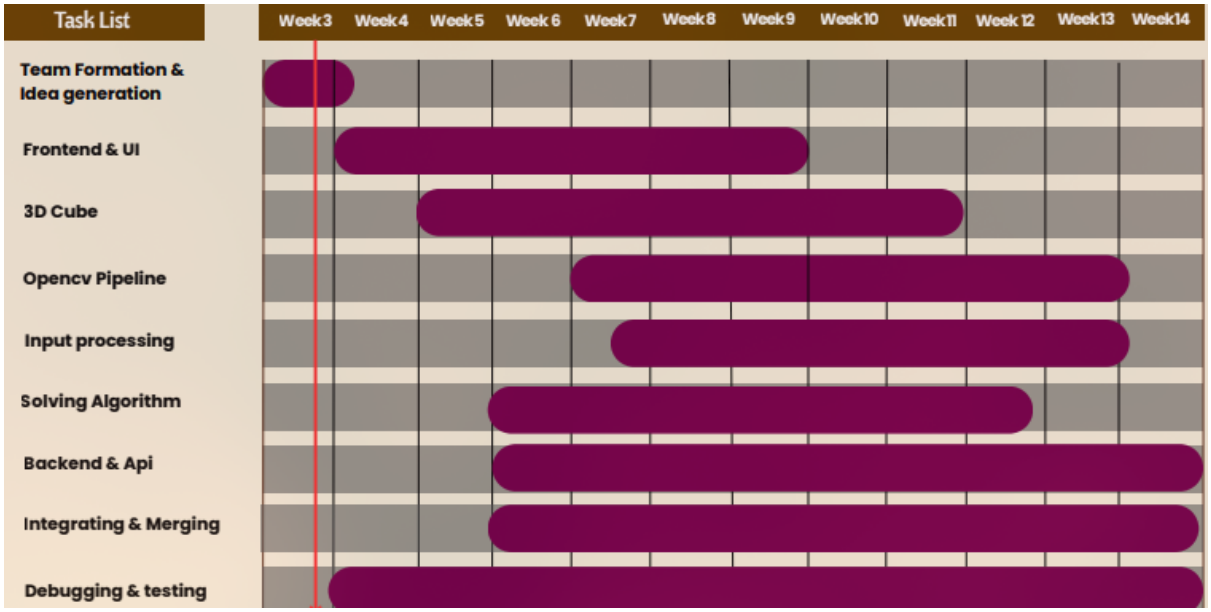


Figure 2: Timeline Diagram (Gantt Chart)

8 Action Taken Based on the Feedback

During our initial review, we received valuable feedback regarding both our development methodology. Below are the specific suggestions and our roadmap to address them.

- Development Strategy: Iterative Implementation

  Suggestion: We were told that theres tons of library and algorithms already built for solving the cube, so building a project like this should not take 3 months, that we should add more features.

  Resolution: We are focusing on the 3x3 cube first. There were already plans to add more features and more cube types, which we will be implementing in the later stages of the project if we finish the 3x3 cube early.

- Timeline?

9 Links

1. Presentation Slide Link
2. GitHub Repository Link